

# Thermal Optimization of a 3U VPX Avionics Cold Plate

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## Abstract

This technical report documents the conjugate heat transfer (CHT) analysis of a 3U VPX standard cold plate. The primary objective was direct: to find the ideal equilibrium between the cooling efficiency of a high-power FPGA and the fluid pumping effort.

## 1 The Problem and Approach

With the continuous increase in power density across aerospace embedded systems, effective heat dissipation has become critical. To address this challenge, I modeled a liquid-cooled cold plate featuring internal micro-channels and simulated the coolant fluid flow within Ansys Fluent.

The numerical simulation utilized the  $k-\omega$  SST turbulence model to precisely capture boundary layer effects. Three distinct mass flow rates (0.01, 0.03, and 0.05 kg/s) were evaluated to analyze how the thermal system dynamically responds to different pumping scenarios in practice.

## 2 Results: Finding the Optimal Operating Point

The post-processed data revealed a distinct behavior of diminishing returns regarding cooling performance. Transitioning the mass flow rate from 0.01 kg/s to 0.03 kg/s yielded a significant and necessary drop in the component's junction temperature. However, further increasing the flow rate to 0.05 kg/s provided negligible additional thermal relief, failing to justify the extra pump power consumption required on the aircraft.

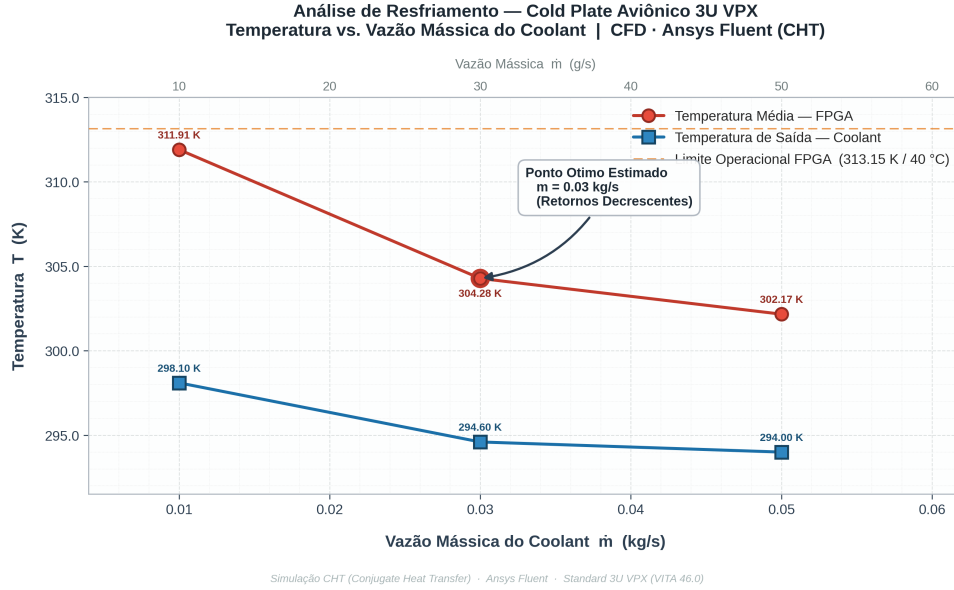


Figure 1: Cooling performance curve generated via Python demonstrating the flattening of thermal optimization benefits past 0.03 kg/s.

Table 1 consolidates the equilibrium values extracted from the solver monitors:

Table 1: Equilibrium Temperatures (Mean FPGA Surface and Fluid Outlet)

| Mass Flow Rate (kg/s) | Mean FPGA Temp (K) | Fluid Outlet Temp (K) |
|-----------------------|--------------------|-----------------------|
| 0.01                  | 311.91             | 298.10                |
| <b>0.03</b>           | <b>304.28</b>      | <b>294.60</b>         |
| 0.05                  | 302.17             | 294.00                |

## 2.1 Thermal Behavior and Flow Characteristics

Visual analysis of the static thermal contours indicates that the heat footprint is heavily concentrated directly at the physical contact interface with the FPGA. Concurrently, the internal coolant fluid successfully performs its role by distributing and dissipating the thermal load across the aluminum chassis structure.

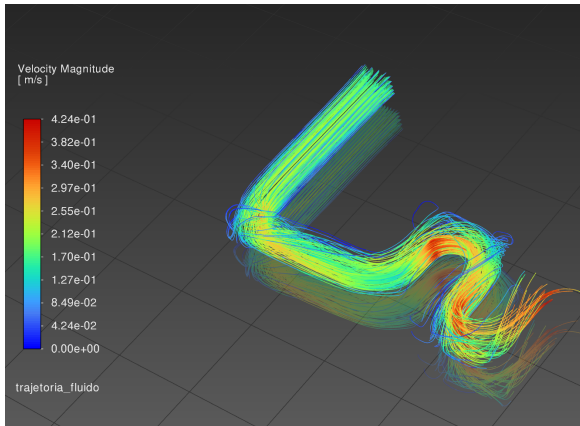


Figure 2: Fluid behavior at 0.01 kg/s (slow flow / higher residence time).

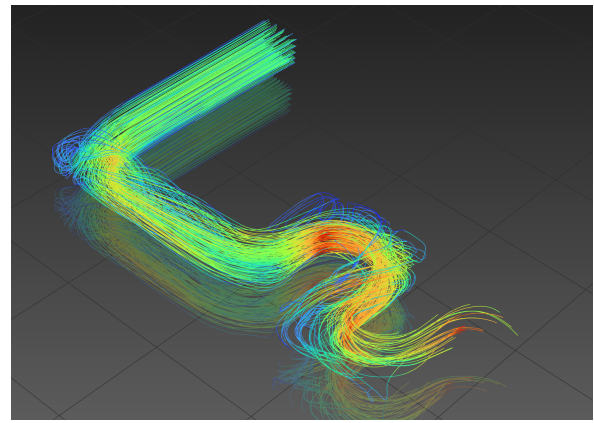


Figure 3: Fluid behavior at 0.05 kg/s (maximum mass flow rate).

Figure 4: Direct comparison of the fluid's thermal profile between the minimum and maximum operating scenarios.

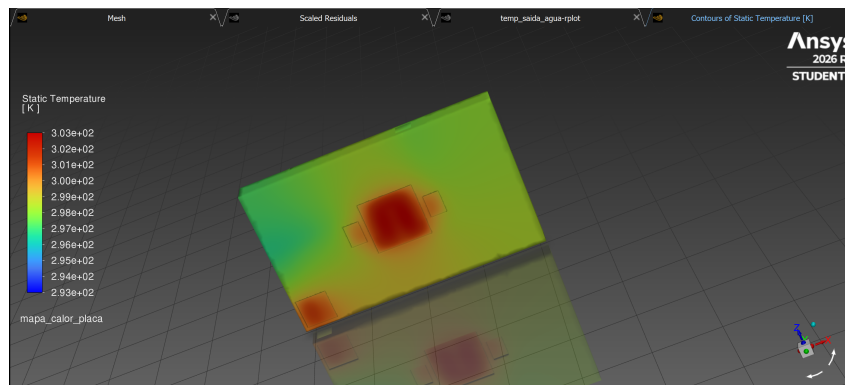


Figure 5: Severe thermal gradient on the bottom surface, highlighting the localized contact zone with the FPGA chip.

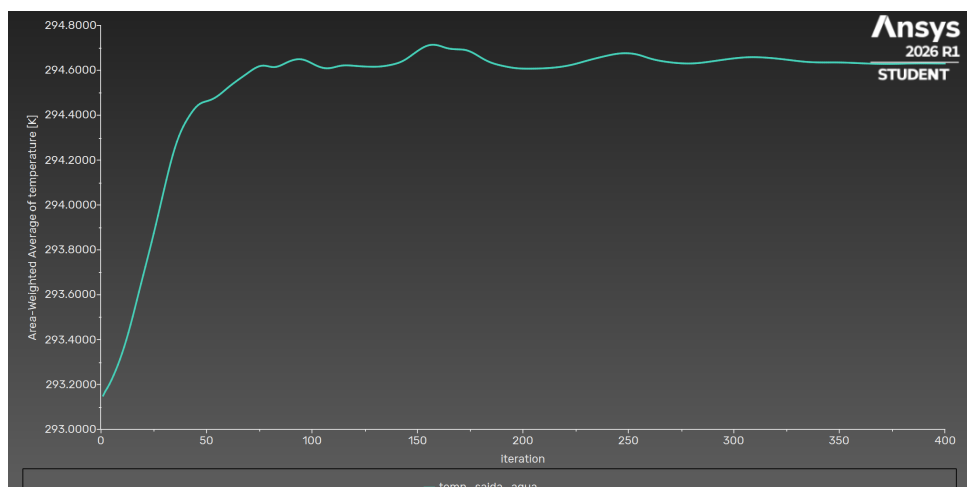


Figure 6: Ansys Fluent convergence monitor demonstrating steady-state stabilization of the fluid outlet temperature at the optimal flow rate.

### 3 Conclusion

The computational study successfully proved that a mass flow rate of **0.03 kg/s** fulfills the rigid safety margins of aerospace embedded systems without over-engineering the project's hydraulic sub-system. This data-driven engineering approach guarantees structural and hardware integrity for real-world avionics applications.